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Intelligence System

[PART ÷ A]

Q1) What is Conditional probability?

→ Conditional probability is the probability of an event occurring given that another event has already occurred. It helps update the likelihood of an event based on available information.

Q2) Explain Bayes' Theorem and its importance in classification.

→ Bayes' Theorem calculates the probability of a class based on observed features. It is widely used in classification because it updates prior beliefs using new evidence and predicts.

the most probable class.

* Importance *

- predicts class membership
- efficient and fast
- works well with High-Dimensional Data.
- Handles Uncertainty
- Basis of Naive Bayes classifier.

(Q3) What assumptions does the Naive Bayes classifier make?

→ Naive Bayes assumes that all features are conditionally independent given the class label. Although this assumption is often unrealistic, the model performs well for many classifications tasks such as spam detection.

(Q84) Explain the working principle of :- ?

→ K-Nearest Neighbours (KNN)

→ KNN is a distance based algorithm that classifies a new sample based on the majority class among its K nearest neighbours. It uses distance metrics such as Euclidean distance.

→ Support Vector Machine (SVM) :-

→ SVM finds an optimal hyperplane that separates classes with the maximum margin. Support vectors are the data points closest to the decision boundary.

Q5) Compare distance based, probabilistic and margin based classifiers?

→ Types :

(1) Distance Based

- KNN
- Uses distance between points.

(2) Probabilistic

- Naive Bayes
- Uses Probability distributions

(3) Margin-Based

- SVM
- Maximizes Separation margin between classes.